Project Design Phase-II
Technology Stack (Architecture & Stack)

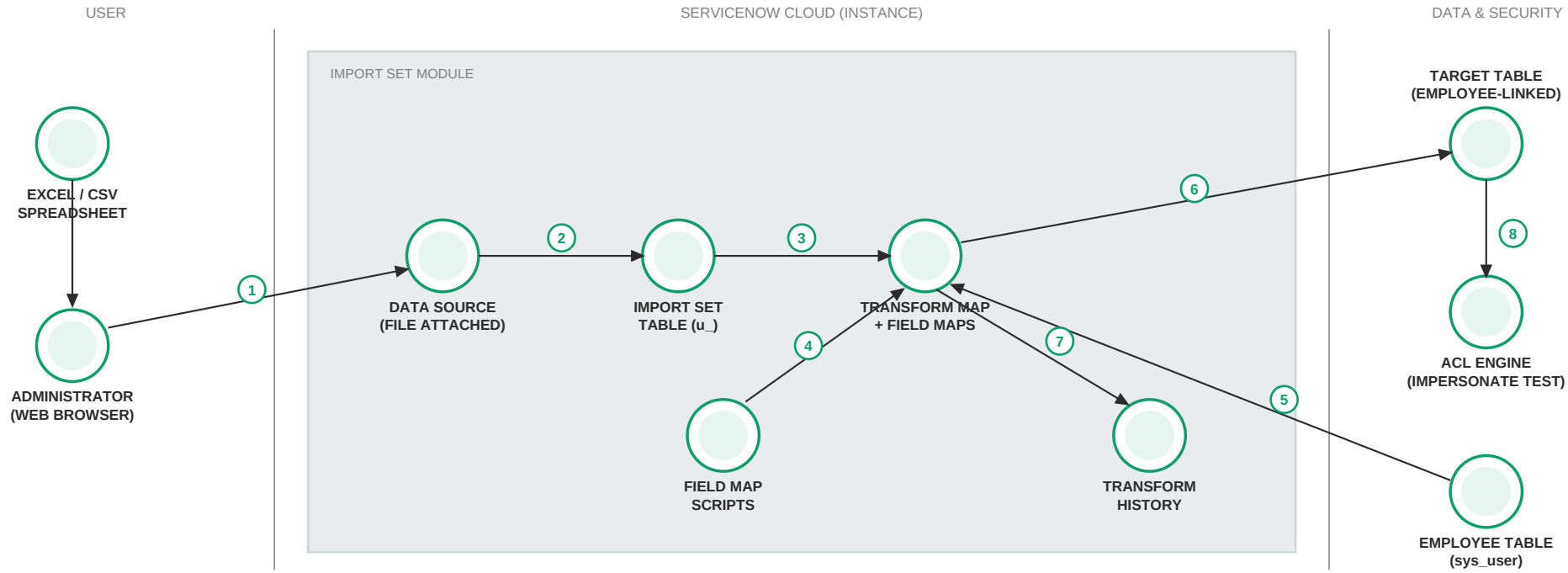
Date	03 October 2026
Team ID	
Project Name	Employee Data Import using Transform Maps (Spreadsheet) - ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Importing employee spreadsheet data using Transform Maps in ServiceNow

Reference: https://docs.servicenow.com (Import Sets and Transform Maps)



Flow: 1 Admin attaches the file to a Data Source; 2 rows are loaded into the staging table; 3 the Transform Map reads the staged rows; 4 field scripts transform values; 5 employee details (department) are looked up from sys_user; 6 linked records are written to the target table; 7 results are logged in Transform History; 8 ACLs secure the target table and are tested with Impersonate User.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Administrator interacts through the ServiceNow web UI (Data Sources, Import Sets, Transform Maps modules)	ServiceNow Next Experience UI (HTML, CSS, JavaScript)
2.	Application Logic-1	Load spreadsheet rows into the staging (Import Set) table	Data Source and Import Set (Excel / CSV)
3.	Application Logic-2	Map staging fields to target fields, link Employee ID to the employee reference and apply Coalesce	Transform Map and Field Maps
4.	Application Logic-3	Pull department and other employee details into the record and transform values	Field map / transform scripts (JavaScript, GlideRecord)
5.	Database	Staging (u_) table and target table schema, fields and reference configuration	ServiceNow tables (relational instance database)
6.	Cloud Database	Employee (sys_user) data, Transform History and Import Log hosted on the cloud instance	ServiceNow cloud instance database
7.	File Storage	Spreadsheet file attached to the Data Source record	ServiceNow Attachments (sys_attachment)
8.	External API-1	Optional: import data through an API instead of a file	REST Import Set API (optional)
9.	External API-2	Optional: bring employee data from an HR / directory system	LDAP / HR system integration (optional)
10.	Machine Learning Model	Not applicable for this project	N/A
11.	Infrastructure (Server / Cloud)	Application deployment on a cloud instance	ServiceNow Cloud (Personal Developer Instance)
	Local Server Configuration:	Web browser and spreadsheet software on the user's computer	Chrome / Edge, Microsoft Excel
	Cloud Server Configuration:	ServiceNow instance with required roles (admin / import_admin)	ServiceNow instance (PDI)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The solution is built on the ServiceNow platform; scripts use standard JavaScript and spreadsheets use open formats (CSV / XLSX)	ServiceNow platform, JavaScript, CSV / XLSX
2.	Security Implementations	Role-based access with ACLs on table and fields, access tested with Impersonate User, authenticated login and encrypted connection	ACLs, Roles, Impersonate User, HTTPS / TLS
3.	Scalable Architecture	Multi-tier cloud platform; the same Transform Map is reused for new files and larger data by loading in batches	ServiceNow multi-tier architecture, reusable Transform Maps
4.	Availability	Hosted on the ServiceNow cloud; imports can be scheduled during off-peak hours	ServiceNow cloud hosting, Scheduled Imports
5.	Performance	Bulk load through a staging table, indexed Coalesce field (Employee ID) and file batching to improve import speed	Import Sets, indexed Employee ID, batch files

References:

https://docs.servicenow.com
https://developer.servicenow.com
https://c4model.com/